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Mapping and Innovating Business Models of Food Hubs in Short Food Supply Chains

Marije Renkema-Singh | Lea Fobbe | Per Hilletofth

Department of Industrial Engineering and Management, University of Gävle, Gävle, Sweden

Correspondence: Marije Renkema-Singh (marije.renkemasingh@hig.se)

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ABSTRACT

Short food supply chains (SFSCs) offer sustainable alternatives to conventional food systems (CFSCs) but require effective intermediaries to scale and remain competitive. Food Hubs (FHs) serve this role by enhancing SFSC efficiency and market reach while upholding community-oriented values. However, limited research has examined how FHs design and innovate their business models (BMs) that enable scaling without compromising their mission-driven goals. This study addresses this gap through an exploratory multiple-case analysis of six Swedish FHs using a BM framework. It provides the first comprehensive empirical investigation of FH BM components, i.e., value intention, proposition, creation and delivery, and capture, and FHs' business model innovation (BMI) strategies. The findings reveal both shared and context-specific BM elements and propose a conceptual framework outlining the core value-related components and BMI strategies of FHs. The study contributes to sustainable SFSC research by offering theoretical and practical insights for designing and scaling resilient, mission-oriented intermediaries in local food systems.

1 | Introduction

The development of sustainable food supply chains (SC) is vital for ensuring resilient food systems that can support current and future generations. At present, the global food supply is predominantly structured around conventional food supply chains (CFSCs), which are typified by long-distance transportation, complex logistics, and reliance on numerous intermediaries. While these systems achieve economies of scale, they are widely criticized for undermining local economies and exerting negative social, environmental, and economic impacts (Zhang et al. 2021). Economically, CFSCs concentrate market power among a few dominant retailers. In 13 EU member states, five major retailers controlled about 60% of food sales (Augère-Granier 2016), limiting small producers' bargaining power and worsening income inequality (Lami et al. 2023). Environmentally, their globalized nature generates substantial ecological costs. The long-distance transport of perishable goods significantly increases greenhouse gas emissions (Marletto and Sillig 2014) and leads to higher food loss and quality degradation (FAO 2017; Rothwell et al. 2016). Socially, CFSCs weaken rural economies by diminishing connections between producers and consumers, reducing transparency and trust, and marginalizing local food systems as industrial-scale operations displace smallholders (Ndimbo and Haulle 2024).

In response, short food supply chains (SFSCs) have emerged as an alternative, offering localized, transparent, and more sustainable approaches to food production, distribution, and consumption (Charatsari et al. 2024; Jarzebowski et al. 2020; Mustapa and Kallas 2025). SFSCs operate within a defined geographical and social framework and mostly include small or medium-sized, family-based enterprises with strong community ties and long-term commitments (Barth et al. 2017; Chiffoleau and Dourian 2020). Their advantages include fostering closer producer and consumer relationships, enhancing

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trust and transparency, and contributing to rural development (Sini 2014; Stoeva et al. 2024). However, SFSCs also face significant challenges, including vulnerability to seasonality, exposure to disruptions, and competition with powerful downstream intermediaries (Sanderson Bellamy et al. 2021). Their reliance on local production makes them sensitive to unpredictable weather conditions and seasonal fluctuations, which can limit product availability and continuity throughout the year (Varecha et al. 2025). Operating on smaller scales, SFSCs often lack the infrastructure and financial resilience needed to withstand shocks such as labor shortages or logistical delays (Aubry and Kebir 2013). Furthermore, actors in SFSCs face structural disadvantages when competing with conventional actors, such as retailers and wholesalers, whose control over pricing, distribution, and consumer access restricts the market opportunities of small producers (Brunori et al. 2016; Lami et al. 2023). For SFSCs to become a viable and scalable alternative to CFSCs, SFSC actors must expand beyond small-scale operations with limited product volumes (Augère-Granier 2016). Research increasingly highlights the importance of intermediaries in this process (Renkema and Hilletofth 2022). Here, alternative intermediaries such as Food Hubs (FHs) are increasingly recognized for their ability to balance economic viability with SFSC values (Renkema-Singh and Hilletofth 2025; Rosol and Barbosa 2021; Tewari et al. 2018).

FHs are mission-driven enterprises that aggregate and distribute locally or regionally produced foods, supporting small and medium producers through product assembly, marketing, and improved market access (Anggraeni et al. 2022; Artoonian et al. 2023; Berti and Mulligan 2016). Unlike conventional intermediaries, FHs operate as social enterprises that combine market efficiency with social objectives, balancing profitability with commitments to local development, environmental sustainability, and community wellbeing (Artoonian et al. 2023; Trivette 2018). Despite their potential, research on the underlying business logic of FHs remains limited. Adopting a business model (BM) perspective allows examination of how FHs propose, create, deliver, and capture value through their strategic configuration of economic and social objectives (Barth et al. 2017; Teece 2010). This perspective is particularly relevant because it provides a holistic understanding of how different activities and stakeholders interact to generate both financial and societal value. The value perspective further enriches this approach by highlighting what kinds of value are created, for whom, and through which mechanisms, beyond purely financial metrics. This is especially important for FHs, where collaboration and value co-creation are the basis for competitive advantage (Renkema-Singh and Hilletofth 2025). Well-designed and continuously innovated BMs enable FHs to coordinate producers, manage logistics, and access markets efficiently while sustaining social and environmental goals (Hyland and Macken-Walsh 2022; Teece 2010). Business model innovation (BMI) thus supports resilience, scalability, and long-term competitiveness, translating the values of SFSCs into viable and sustainable strategies, which is an area still underexplored in the agri-food sector (Donner and de Vries 2023; Geissdoerfer et al. 2018).

Much of the existing literature on FHs focuses on descriptive aspects, emphasizing their operational capabilities and relational capabilities (Renkema-Singh and Hilletofth 2025), while overlooking the logic of how they operate and innovate. When SFSC literature does investigate BMs and to some extent BMI, they do not specifically look into FHs. For example, Donner and de Vries (2023) identify sustainable BM types across food systems, recognizing FHs as part of these systems, but without examining the distinct BM components or discussing BMI of FHs. Similarly, Manikas et al. (2019) and Hyland and Macken-Walsh (2022) classify FHs into model types but provide limited insights into their underlying logics and innovation strategies to uphold and strengthen competitive positioning in the market. Prior studies often highlight social and economic value creation and delivery in FH operations (Artoonian et al. 2023; Saul et al. 2021), yet other crucial BM dimensions, including value intention, propositions, capture, and innovation strategies through BMI, remain underexplored. Consequently, there is insufficient knowledge about how FHs design and innovate their BMs in ways that allow them to scale while maintaining their mission-driven values.

Addressing this research gap requires a systematic investigation of FH BMs and the innovation strategies they employ to grow and sustain operations. A comprehensive understanding of FH BMs and BMI would not only advance academic theory but also provide practical guidance for managers and policymakers seeking to strengthen SFSCs. To address this gap, this study poses the following research questions:

- How can Food Hubs' business model components be described?
- Which strategies can be identified as contributing to FH's business model innovation?

To answer these questions, the BMs and BMI of six FHs operating in the Swedish market were explored.

The contributions of this research are threefold. First, it extends SFSC scholarship by moving beyond producer-centered perspectives and offering a detailed account of an intermediary as a critical scaling actor. Second, it advances BM research in the agri-food sector by proposing a conceptual BM framework for FHs, detailing aspects such as value proposition, value creation and delivery, and value capture. It also integrates entrepreneurial value intentions and considers the innovation strategies used by FHs in their BMI. Third, while collaboration and relationships are widely recognized as central to FH operations, prior work has not systematically mapped FHs' BM components and BMI that incorporate these aspects, nor presented them in a way that allows for replication by researchers and practitioners. This study presents actionable insights for practitioners, illustrating how FHs can design and innovate their BMs to achieve sustainable competitive advantage while scaling their operations.

The remainder of the paper is structured as follows. Section 2 provides a theoretical framework based on BM and BMI literature and reviews BM and BMI research in SFSCs and FH literature. Section 3 outlines the methodological approach used in the study. Section 4 presents the results, focusing on the BM components and BMI strategies identified across the six FHs. Section 5 discusses how the innovation strategies affect the different BM

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components and relate to BMI, and Section 6 concludes with answers to the research questions, discusses implications of these findings for both theory and practice, and finally discusses avenues for further research.

2 | Literature Review

2.1 | Business Models

BMs function as strategic blueprints that describe how firms propose, create, deliver, and capture value. Typically structured around three components, namely value proposition, value creation and delivery, and value capture, BMs outline what a company offers, how it differentiates itself from competitors, the processes and resources involved, and how financial and non-financial returns are generated (Teece 2010).

As societal demands for sustainability intensify, both researchers and practitioners have extended the traditional BM framework toward more sustainable configurations by incorporating economic, environmental, and social sustainability dimensions into each of its components (Schaltegger et al. 2016). Central to this approach is the co-creation of value with a wide network of stakeholders, including employees, suppliers, local communities, and society as a whole (Attanasio et al. 2022; Fobbe et al. 2024; Upward and Jones 2016). This perspective shifts the emphasis from a narrow focus on profit maximization to the broader pursuit of shared value creation (Porter and Kramer 2011). Given the growing consensus among scholars that sustainable BMs confer competitive advantage, the sustainable BM concept is increasingly regarded as a successor to the conventional BM paradigm (Geissdoerfer et al. 2018). Accordingly, in the context of this article, the term 'BM' is used to denote models that inherently integrate sustainability indicators.

Some scholars identify value intention as an additional BM component in the agricultural sector, reflecting owners' commitment to innovation and sustainable value creation. As many agricultural firms are family-run and deeply rooted in their communities, owners often assume a stewardship role, managing their businesses with a sense of responsibility toward both economic performance and environmental and social well-being (Barth et al. 2017). Through stewardship, business leaders translate their values into concrete BM choices that balance profitability with care for the environment, community, and future generations (Ulvenblad 2023). In developing sustainable food supply chains, the integration of sustainability indicators, organizational values, and continuous innovation should be aligned with the owner's vision and value intention, acting as key elements shaping BM development (Attanasio et al. 2022; Donner and de Vries 2023). While some scholars view stewardship as one component within the BM, others conceptualize it as a higher-order principle, a driving force behind sustainable BMI (Ulvenblad 2023).

Finally, maintaining profitability and enhancing sustainability requires companies to remain agile by continuously innovating and refining their existing BMs. This process, known as BMI, involves applying various innovation strategies that adapt to or transform one or more BM components (Geissdoerfer et al. 2018). The choice of type of strategies involving BMI depends greatly on the internal and external changes affecting a company's existing BM. Such changes may take the form of minor, incremental adjustments or, alternatively, major transformations often referred to as radical changes (Boons and Lüdeke-Freund 2013). Geissdoerfer et al. (2018) discuss four different BMI types which are sustainable start-ups, diversification, transformation, and acquisition. Each of these innovation types reflects a distinct pathway through which organizations can evolve their BMs to enhance sustainability and competitiveness. Sustainable start-ups refer to the creation of a new organization with a new BM. BM diversification is the process in which companies develop additional BMs, allowing them to experiment with BMs without disrupting their core operations. It supports learning and gradual shifts toward sustainability while exploring new markets or stakeholder needs. BM transformation occurs when companies modify their existing BM to improve their sustainability performance or align it with new strategic goals. And finally, BM acquisition occurs when an additional BM is identified, acquired, and integrated into the organization (Geissdoerfer et al. 2018) (Figure 1).

2.2 | Short Food Supply Chains

In order to understand the context in which FHs operate and develop and innovate their BMs, it is important to understand SFSCs. SFSCs are social supply chains built on stakeholder collaboration and inclusivity, pursuing economic, environmental, and social sustainability (Brunori et al. 2016; Stoeva et al. 2024). Their value creation is rooted in trust, authenticity, and locality, distinguishing them from CFSCs (Sebök et al. 2022). Their perceived benefits include reduced CO_2 emissions, stronger rural–urban links, job creation, community building, and cultural support (Galli and Brunori 2013; Lulovicova and Bouissou 2024). SFSCs operate as direct-to-consumer or intermediate models, with the latter offering greater scalability and competitiveness (Renkema and Hilletofth 2022).

Beyond tangible resources, SFSC stakeholders rely on intangible assets like relationships, knowledge, and collaboration for BM differentiation (Charatsari et al. 2023). To address constraints such as small volumes, logistics, and market access, many companies operating in SFSCs adopt hybrid distribution models combining direct sales with sales to business customers, including retail and public procurement, with the goal to improve turnover and strengthen local food systems (Aggestam et al. 2017). Firms in SFSC often face challenges including resource limits, external crises, and changing consumer behavior, requiring BMI (Charatsari et al. 2023; Csordás et al. 2022). Through BMI, SFSCs can develop new mechanisms for BM components, and such innovations can be technological and non-technological and include increased digitalization, product development, and new marketing and cooperative structures, with the goal to enhance operations, authenticity, traceability, and trust (Sebök et al. 2022). Especially digitalization has been recognized as a key BMI driver, helping firms optimize operations and boost resilience despite initial resource gaps (Csordás et al. 2022). Finally, diversified sales channels and strong stakeholder collaboration are recognized as being important in accomplishing BMI

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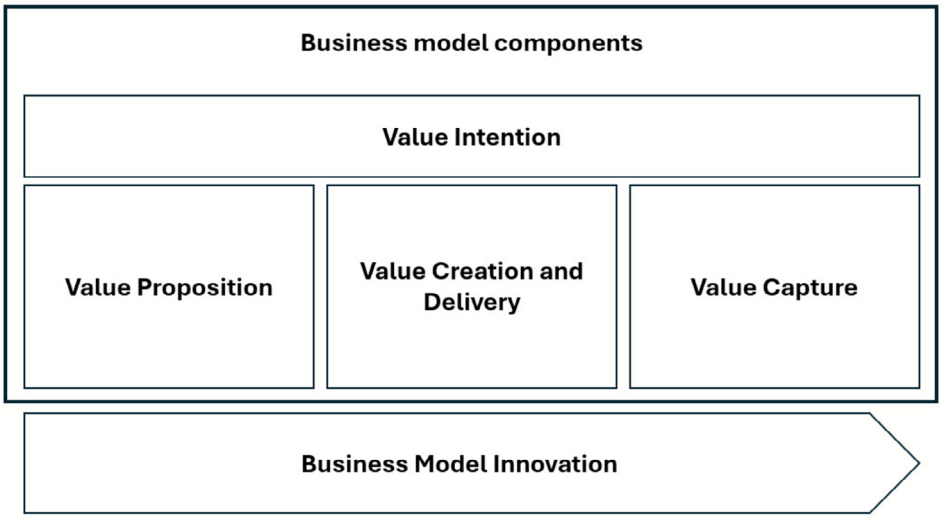


FIGURE 1 | Business model components and business model innovation framework own elaboration based on (Barth et al. 2017; Bocken et al. 2014; Geissdoerfer et al. 2018).

(Keech et al. 2023). Understanding BMI helps identify how SFSCs can scale without compromising their mission-driven values, ensuring both economic viability and long-term societal impact.

2.3 | Food Hubs

Within SFSCs, FHs can be a driving intermediary creating value for their own organizations as well as SFSC stakeholders by performing critical functions within SFSCs, including increased market access for local products (Christensen et al. 2019; Renkema-Singh and Hilletofth 2025). There is still limited knowledge regarding FHs' specific BM and BMI strategies, but they are commonly classified into three operational typologies: direct-to-consumer FHs, producer-to-business FHs, and hybrid FHs (Hyland and Macken-Walsh 2022). Direct-to-consumer FHs facilitate access to local products through targeted marketing to consumers, whereas producer-to-business FHs connect producers to larger institutional buyers such as hotels, restaurants, and catering businesses, retailers, and public institutions. Hybrid FHs combine both market channels, employing diversified strategies to reach a wider customer base (Hyland and Macken-Walsh 2022).

In the literature, FHs are typically framed around two primary value propositions, namely a value-based orientation focused on scaling local food systems, improving profitability, strengthening distribution networks, and enhancing producer negotiation power; and a sustainability-focused approach emphasizing social equity, environmental responsibility, and public health (Manikas et al. 2019). They prioritize long-term partnerships with local producers, generate revenue through aggregated sales and service provision, and simultaneously enhance the social and environmental resilience of the food system (Jia et al. 2024; Sgroi and Marino 2022; Tewari et al. 2018). In this way, FHs integrate business and social sustainability objectives, creating economic returns while advancing broader community goals (Artoonian et al. 2023). Thus, FHs have the capability to reinforce the competitiveness and resilience of SFSCs, making them central intermediaries potentially enhancing the sustainability of SFSC stakeholders (Anggraeni et al. 2022; Artoonian et al. 2023). However, in order to be successful, scholars stress the importance of aligning both these business and social sustainability objectives in FH BMs (Saul et al. 2021).

Irrespective of their market focus, FH's aggregate and assemble locally sourced products, ensuring that producers can reach diverse market segments (Anggraeni et al. 2022; Artoonian et al. 2023; Berti and Mulligan 2016; Matson and Thayer 2013). They often commit to strengthening community ties, supporting local producers, and improving access to healthy, locally sourced food (Artoonian et al. 2023). FHs develop and manage distribution channels, ranging from retail outlets to public procurement schemes. By coordinating logistics, facilitating joint marketing, and supporting horizontal collaboration among producers, FHs enable small-scale enterprises to collectively meet the demands of larger customers such as wholesalers and institutional buyers (Engelseth 2016; Hyland and Macken-Walsh 2022; Rogers and Fraszczak 2014). This collaborative approach improves operational efficiency and strengthens relationships between stakeholders in SFSCs, strengthening these initiatives (Engelseth 2016; Hyland and Macken-Walsh 2022). Moreover, through branding, packaging, and marketing of source-identified products, FHs enhance product differentiation, visibility, and consumer trust (Barham et al. 2012).

3 | Research Method

Despite the knowledge regarding market focus, orientation, and operational capabilities such as logistics and marketing (Renkema-Singh and Hilletofth 2025; Schmit and Severson 2019), empirical research employing a comprehensive BM perspective, investigating all BM components and their interdependencies among components and the roles of value intention and BMI, is still underdeveloped in the literature. Thus, this study adopts an exploratory multiple case study approach, allowing for the exploration of the phenomenon of the FH

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within its real-world context, drawing on diverse data sources to bridge theory with empirical findings (Piekkari et al. 2009).

3.1 | Case Selection

Six FHs in Sweden have been investigated in this study. The cases were chosen by pursuing a literal replication logic (Yin 2018). Although the term "Food Hub" is more commonly used in North America, the researchers intentionally selected cases that align with the widely accepted definition found in the literature. Specifically, the cases chosen for this study share key characteristics with FHs as defined in the context of SFSCs, that is: intermediaries responsible for aggregating and selling locally produced goods to domestic customers, using either their own distribution channels or through strategic partnerships. Investigating FHs in Sweden is especially relevant due to the country's strong alignment with sustainability policies and increasing public interest in local food systems. This commitment is shown through the development of national food strategies and reflected in its public food procurement strategies (Regeringskansliet 2025). While Swedish consumers appear to be conscious of sustainability issues, and local producers are interested in selling through SFSCs, there is little research conducted on how these local networks can develop with FHs as an important intermediary. This presents a unique opportunity to examine this intermediary and its BM.

To enhance external validity, the six FH cases were selected from diverse geographical regions in Sweden and represent varying market orientations (Table 1). All FHs are small companies where the founder is often also the CEO and is actively involved in all parts of the FH operations. FHs A–E focus on the sales of exclusively regional products, whereas FH F also includes national products. They sell to mainly business customers (FH B–F) or consumers (FH A). To confirm the suitability of these cases, the FHs were approached via e-mail and telephone before setting up meetings for company visits and interviews. These phone calls also helped to obtain more information about the different experts that might be willing to be interviewed for this study. All participation was voluntary, and no remuneration of any kind has been paid to the companies or participants of this study.

3.2 | Data Collection

The data has been collected between 2022 and 2025, a period with SC disruptions caused by the pandemic and geopolitical unrest affecting agricultural business all over the world. To ensure construct validity of this study, the researchers used multiple sources of evidence (Yin 2018). The whole case study consists of 22 semi-structured interviews conducted with 16 employees in different positions in the case companies (Table 2). In addition, the researchers conducted observations of the companies' offices, warehouses, and other operations and gathered documentation such as marketing communications with suppliers and customers. Finally, during these years, the researchers followed the case companies by follow-up interviews and retrieving articles appearing in mass media and consulting archival records such as annual financial reports.

TABLE 1 | Information on case studies.

Case study	FH A	FH B	FH C	FH D	FH E	FH F
No. employees (2023)	8	24	8	9	7	14
Established	2018	2008	2017	2015	2015	1995
Location	South Sweden	South Sweden	East Sweden	West Sweden	West Sweden	East Sweden
Main customers	Consumer	Hotel, Restaurant Catering	Hotel, Restaurant Catering	Hotel, Restaurant Catering	Public procurement	Hotel, Restaurant Catering
Remaining customers	Public procurement	Public procurement	Public procurement	Public procurement	Hotel, Restaurant Catering	Public procurement
	Hotel, Restaurant Catering	Consumers			Retail	Retail
Gross turnover	10,033,000 (2022)	76,485,000 (2022)	30,259,000 (2023)	29,285,000 (2023)	38,340,000 (2023)	111,954,000 (2022)

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The conducted interviews were semi-structured where the researchers followed a guide which included the topics and issues to be studied (Easterby-Smith et al. 2021). To enhance the reliability of the study, the researchers developed a case study protocol designed around a few core themes and maintained a comprehensive case study database (Yin 2018). The core themes included a focus on the company's BM and innovation strategies, as well as a focus on stakeholder engagement and sustainability. In this way, the interviewee was free to explain their knowledge and experience, without any predefined construct or theory. This is important, as the concepts or terms described by academic literature are not necessarily the terms that the interviewees would use in their description (Gioia et al. 2013). The interviews with the CEOs or operative managers explored mainly strategic areas and business operations while expert employees were interviewed based on their specific area of expertise. FH managers were asked about their key drivers for business development and for maintaining competitiveness or market relevance. All interviews took between 30 and 90 min. All processed data was discussed with the case companies to confirm an accurate depiction of the results. Furthermore, the findings presented in this paper were shared with the case companies to ensure transparency and credibility.

TABLE 2 | Conducted interviews.

Case study	Interviewee	Year	Time (min)
FH A	CEO/Founder	2022, 2025	60, 20
	Marketing manager	2022, 2023	30, 30
	Consumer sales manager	2022	60
FH B	CEO/Founder	2022, 2025	90, 60
	Purchasing, sales, and producer relations	2022	30
	Warehouse-/ Retail manager	2022	30
	Public Procurement sales	2022	30
FH C	CEO/Founder	2023	30
FH D	CEO/Founder	2023	30
	Sales manager/ Founder	2023	30
FH E	CEO/Founder	2022	30, 60
	Co-owner/Sales manager	2022	60
FH F	Office manager	2023, 2025	90, 60
	Retail sales	2023	30
	Warehouse manager	2023	30
	Logistics manager	2025, 2025	30, 30

3.3 | Data Analysis

To ensure strong internal validity, the researchers employed a multi-phase analytical strategy (Yin 2018). The gathered data was structured and analyzed using Gioia-inspired coding structures (Figure 2) (Gioia et al. 2013). In the lower levels of the coding scheme, the researchers have taken an inductive approach to analyze the data, involving open coding through a full review of the dataset and the identification of key sentences and concepts, grouping them according to recurring themes and distinct differences. This was followed by axial coding, in which the data was organized using the developed BM framework (Figure 1). At this stage, the analysis shifted to a more deductive orientation, allowing empirical findings to be systematically examined through the framework. This facilitated the identification and interpretation of components that define FH BMs and innovation strategies affecting BMI. To further strengthen external validity, theory-based generalization was employed to develop these BMs and innovation strategies reflective of the FHs studied. The structured data

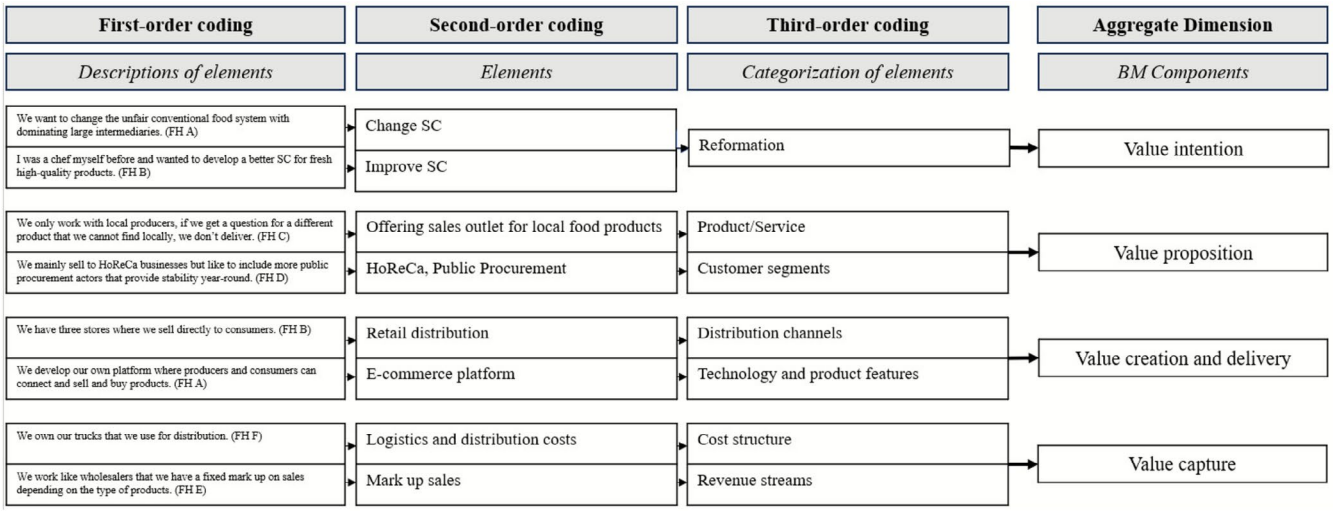


FIGURE 2 | Example of data analysis coding.

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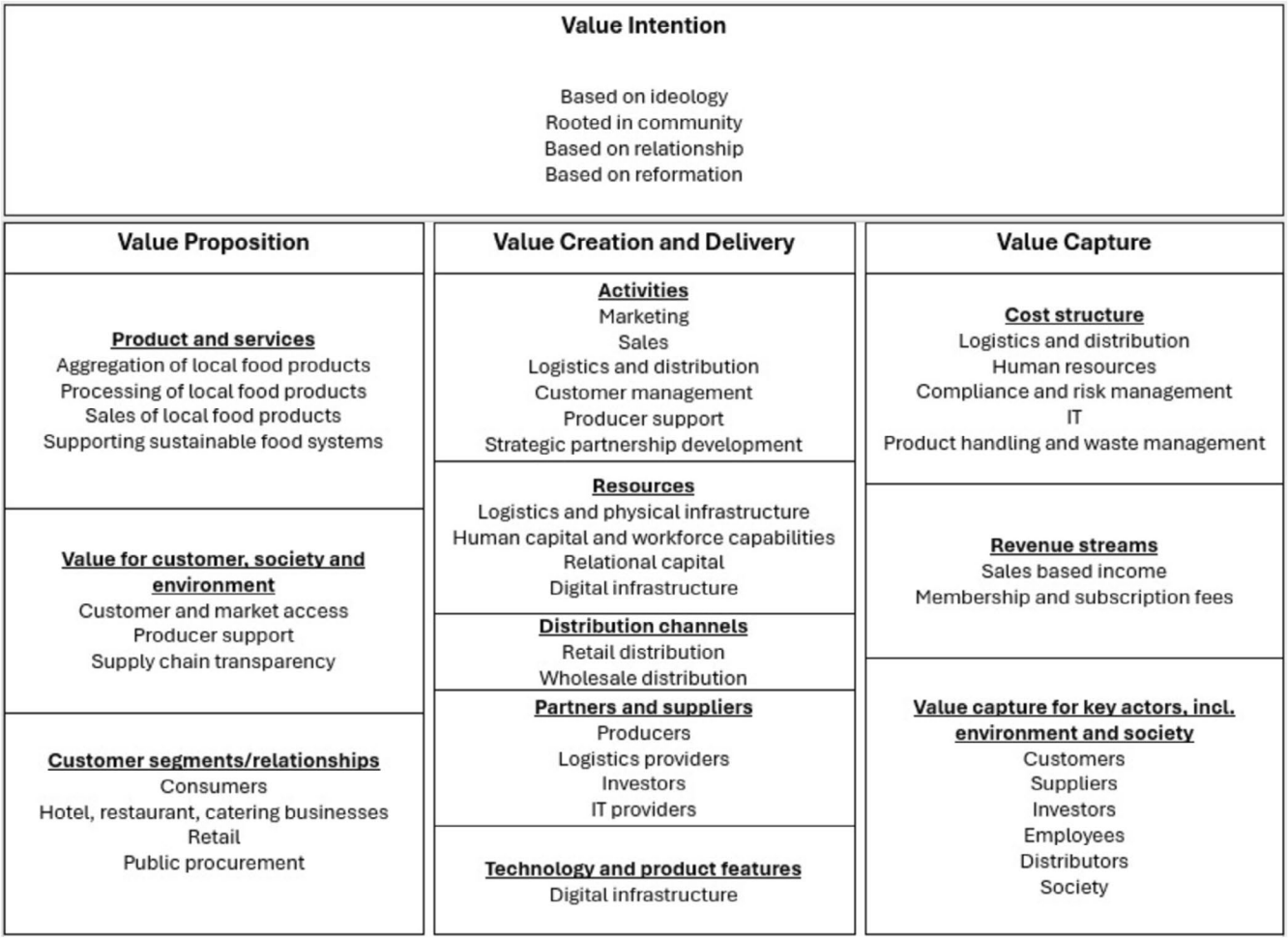


FIGURE 3 | Business model components in FHs (*Source:* Author's own elaboration).

enabled a within-case analysis of each FH, focusing on components such as value intention, value proposition, value creation and delivery, and value capture.

Finally, a cross-case analysis was conducted to identify both convergent and divergent patterns across the cases, contributing to the formulation of a conceptual BM and identification of strategies for BMI being tailored to FHs in the context of sustainable food systems (Figure 2).

4 | Results

This section explores the BM components of FHs and the different innovation strategies affecting the BMI efforts of the FHs.

4.1 | BM Components in FHs

The results showed that FHs have multidimensional and complex BM with several elements in the different BM components (Figure 3). The text under each subchapter describes the key points, while the tables give a full overview of all identified elements.

4.1.1 | Value Intention of Food Hubs

All the FHs managers explained that their vision for the company is driven by passion and ideology. They emphasize fairness, transparency, and shared value among all stakeholders (Table 3). Their motivation extends beyond economic gain; they view their work as part of a broader mission to strengthen local food systems and ensure that value is equitably distributed across the supply chain. The managers' personal and professional backgrounds reinforce this stewardship role. FH A's manager is a producer, FHs B–D have founders with long experience in the hospitality sector, FH E's manager previously worked in the agri-food industry, and FH F's operational manager comes from the retail sector. These embedded experiences create a strong sense of belonging and accountability to the regions in which they operate. Consequently, the FH managers explained their perception of producers as partners and customers as collaborators, rather than transactional counterparts. Because of their close relations with both producers and customers, they are the first to detect and respond quickly to emerging challenges, such as supply disruptions or shifts in demand. Moreover, they see themselves and their companies as part of a symbiotic food system, where SC stakeholders collaborate closely, sell good products, give good service, and collectively pursue mutual economic benefit.

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The value intention of FH A and FH B differs slightly from that of the other FHs, as both aspire not only to distribute food but also to act as reformers within the food system. Their mission extends beyond commerce, adopting an activist stance aimed at challenging the prevailing lack of transparency in CFSCs. In interviews, the CEO of FH B described his entrepreneurial vision as one of systemic change, aiming to build a supply chain that benefits the entire local economy rather than serving the interests of a single dominant actor.

4.1.2 | Value Proposition of Food Hubs

Across all studied FHs, the value proposition centers on providing consumers with local, fresh, and high-quality food products (Table 4). The notion of local is context-specific and for FHs B–F, it typically refers to food sourced and sold within regional or provincial boundaries, reinforcing their commitment to strengthening local food economies. These FHs act primarily as wholesalers, aggregating products from nearby producers and distributing them within the same geographic area to public procurement clients and HoReCa customers. By contrast, FH A currently sources products from a specific region and distributes nationally. However, its strategic vision is to create place-based connections by linking producers and consumers within shared regions across Sweden. Rather than aggregating products, FH A facilitates direct exchange through its e-commerce platform, positioning localness as both a logistical and relational aspect of its value proposition.

Further, the FHs emphasize long-term relationships with both producers and customers, grounded in quality, fairness, and trust as a crucial part of their value proposition. For producers, FH offers fair pricing, stable market access, and enhanced competitiveness, and for customers, they offer high-quality products, traceability, and ethically sourced local products. Thus,

TABLE 3 | Value intention of FHs.

	FH A	FH B	FH C	FH D	FH E	FH F
Value intention						
Based on ideology	X	X	X	X	X	X
Rooted in community	X	X	X	X	X	X
Based on relationship	X	X	X	X	X	X
Based on reformation	X	X				

TABLE 4 | Value proposition components of food hubs.

	FH A	FH B	FH C	FH D	FH E	FH F
BM value proposition						
Product/service						
Offering a sales outlet for local food products		X	X	X	X	X
Processing primary products adding shelf life, consistency, and customer appeal while reducing waste and production costs.		X	X			
Enabling market access for producers	X	X	X	X	X	X
Offering new marketing outlets, while supporting producer autonomy	X					
Value for customer, society and environment						
Offering local food to local businesses and consumers.	X	X	X	X	X	X
Supporting local food production	X	X	X	X	X	X
Providing support for local producers	X	X	X	X	X	X
Increasing social proximity by enabling transparency and communication in the supply chain	X	X	X	X	X	X
Customer segments/relationship						
Consumers	X	X				
Hotel, restaurant, catering businesses		X	X	X	X	X
Public procurement	X	X	X	X	X	X
Retail						X

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their close collaboration with producers and customers reflects a relational value proposition that prioritizes transparency and mutual benefit over purely transactional exchanges. Through continuous dialog, regular visits, and coordination of supply and demand, FHs ensure reliability and responsiveness, which are key elements of the value they promise to both sides of the chain. By integrating economic and social objectives, FHs transform relationships themselves into a core component of their value proposition.

The customer segments define for whom value is created and include the needs, preferences, and problems of the customer the FH is trying to solve. In this study, FHs served a range of segments, including individual consumers, HoReCa businesses, public procurement entities, and retail stores. Among the six FHs analyzed, two (FH A and B) adopted a hybrid model, serving both private consumers and business customers, while the remaining four (FH C–F) focused exclusively on business clients, including public procurement. The selection of customer segments reflected each FH's contextual conditions and strategic orientation. FH A positioned itself as a business-to-consumer intermediary, targeting individual consumers. FHs B, C, and D concentrated on HoReCa clients, drawing on the founders' professional experience and industry networks. FH E prioritized public procurement to secure stable, year-round demand. FH F, located in a tourism-driven area, targeted HoReCa businesses as its main customer base, though seasonally fluctuating.

4.1.3 | Value Creation and Delivery of Food Hubs

The value creation and delivery components of FHs are sorted into activities, resources, distribution channels, partners, suppliers, and technology and product features (Table 5). The activities illustrate the coordinated processes through which FHs meet customer needs. Similar to conventional wholesalers, the FHs create value primarily through logistics and distribution solutions, either by operating their own delivery systems or partnering with third-party logistics providers. FHs B and C also engage in small-scale processing, where FH B produces bread, ready meals, and offers catering, and FH C processes meat through smoking and hanging. FHs A and B additionally target end consumers via retail channels, operating either a physical shop (FH B) or collaborating with a distributor for home deliveries (FH A).

Key resources include both tangible assets (warehouses, vehicles, and logistics infrastructure) and intangible assets, notably a skilled and loyal workforce. In FHs B, C, and D, staff with extensive HoReCa experience brought valuable industry knowledge that shaped service quality and responsiveness, allowing precise alignment with customer needs. Personal networks developed through prior professional experience also enhanced customer retention and acquisition. In an industry with high staff turnover, these networks generated continuity as chefs and clients often maintained supplier relationships when changing workplaces, contributing to the organic growth of FHs B and D.

All FHs emphasized trust-based partnerships as central to value creation and delivery. Their ongoing interactions with producers and customers represent relational capital, enhancing coordination and operational efficiency. Producers are treated not merely as suppliers but as co-creators of value, collaborating closely to maintain responsiveness, quality, and transparency. Continuous communication helps balance supply and demand, ensuring timely delivery and reducing inefficiencies. This dynamic coordination supports long-term value creation, such as helping producers diversify crops (FH C), redirect surplus to new buyers (FH E), and minimize waste (FH B), thereby strengthening both economic and environmental sustainability across the SFSC.

Value creation and delivery among the FHs is partly shaped by their technological infrastructure. FH A, operating as an e-commerce company, invests heavily in its digital platform, which functions as the primary interface for connecting suppliers and customers. In contrast, the other FHs show lower digital maturity, relying on basic tools to support operational functions, such as standard ordering platforms, Excel spreadsheets, or SMS and phone orders. FHs B and E noted that these limitations often stem from small-scale producers' lack of digital skills, financial capacity, or motivation to adopt advanced systems, which are factors that influence the overall efficiency and smoothness of value delivery across the supply chain.

All FHs pursue a dual objective of minimizing lead times while accommodating flexible delivery schedules. Aligned with their value proposition of providing fresh, high-quality, locally sourced food, they avoid large inventories and rely on on-demand delivery models with short intervals between harvest and distribution to reduce food waste. When surpluses occur, FHs apply targeted waste management strategies, such as in-house processing (FHs B and C) or redirecting unsold products to biogas plants (FHs E and F), thereby maintaining efficiency and environmental responsibility.

4.1.4 | Value Capture of Food Hubs

The main fixed costs across all FHs relate to warehouse rent, logistics and distribution (via owned or third-party vehicles), product aggregation and handling, staff salaries, and administrative expenses such as IT and insurance. Revenue streams primarily come from product and service markups (FHs B–F). In contrast, FH A, which does not aggregate products, operates on a commission-based model, earning a fee on sales and charging producers a subscription for use of its e-commerce platform. Subscription packages vary by price and service level, offering different degrees of marketing, exposure, and sales support (Table 6).

All FHs emphasized that value is captured collectively by multiple stakeholders, including customers, suppliers, employees, distributors, investors (FH A), and the broader community. Customers benefit from access to high-quality, locally sourced products that enhance freshness, transparency, and trust. Suppliers gain from fair and stable pricing, supporting economic resilience and market positioning. For employees, participation in socially and environmentally driven organizations fosters loyalty, job satisfaction, and commitment (FH B).

Value is also captured for society and the environment through increased local food sales, rural development, job creation, and reduced food waste, contributing to sustainability

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TABLE 5 | Value creation and delivery components of FHs.

	FH A	FH B	FH C	FH D	FH E	FH F
BM value creation and delivery						
Activities						
Sales	X	X	X	X	X	X
Processing		X	X			
Logistics and distribution	X	X	X	X	X	X
Sourcing and procurement	X	X	X	X	X	X
Marketing	X	X	X	X	X	X
Customer relationship management	X	X	X	X	X	X
Revenue management	X	X	X	X	X	X
Strategic SC partnerships development		X	X	X	X	X
Resources						
Warehouse	X	X	X	X	X	X
Logistics and distribution	X	X	X	X	X	X
Digital platform	X					
Close relationships with suppliers	X	X	X	X	X	X
Close relationships with customers		X	X	X		X
Skilled workforce		X	X	X	X	X
Strong company culture		X				
Distribution channels						
Retail distribution	X	X				
Wholesale distribution		X	X	X	X	X
Partners and suppliers						
Local producers	X	X	X	X	X	X
Third party logistics providers	X				X	
IT providers	X					X
Investors	X					
Technology and product features						
E-commerce platform	X					
IT systems for stock and order management and invoicing	X	X	X	X	X	X
IT systems for logistic solutions						X

and food security. Close relationships and rapid communication among FHs, producers, and customers further improve SFSC efficiency. Third-party logistics providers capture value through delivery fees, while in FH A's case, they also gain reputational benefits by supporting sustainable local supply chains. FH E enhances financial and environmental efficiency by co-locating with a conventional wholesaler, enabling truck sharing that reduces both costs and carbon emissions. Lastly, FH A's venture capital partner captures value through returns on sustainable revenue growth and the FH's expanding market reach.

4.2 | Innovation Strategies of Food Hubs

The study identified four different strategies that are used by the different FHs to innovate their BM (Table 7).

4.2.1 | Digitalization Strategy Enhancing Efficiency and Scalability

While there is a growing awareness of the potential for digital tools to enhance efficiency and scalability, particularly in managing

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TABLE 6 | Value capture components of food hubs.

	FH A	FH B	FH C	FH D	FH E	FH F
BM value capture						
Cost structure						
Warehouse	X	X	X	X	X	X
Logistics and distribution costs	X	X	X	X	X	X
Salaries staff (fixed and variable)	X	X	X	X	X	X
Insurances	X					X
IT costs	X	X	X	X	X	X
Platform development costs	X	X	X	X	X	X
Certification costs	X	X	X	X	X	X
Waste management	X	X	X	X	X	X
Customer relationship management	X	X	X	X	X	X
Revenue management	X	X	X	X	X	X
Strategic SC partnerships development		X	X	X	X	X
Revenue streams						
Commission on sold products	X					
Subscription fees	X					
Mark up sales		X	X	X	X	X
Value capture key actors including environment and society						
Customers: enjoy high-quality, fresh food from local producers	X	X	X	X	X	X
Suppliers: selling products for a fair price, increasing profits and strengthening competitive advantage	X	X	X	X	X	X
Investors: ROI	X					
Employees: salaries, skills development, job satisfaction	X	X	X	X	X	X
Distributors: distribution fee, positive PR	X					
Society: increased sales from local food products, enabling rural growth, local jobs, food security, less food waste due to freshly harvest food	X	X	X	X	X	X

TABLE 7 | Business model innovation strategies.

	FH A	FH B	FH C	FH D	FH E	FH F
Business model innovation						
Innovation strategies						
Digitalization strategy	X					X
Diversification strategy		X				
Collaboration strategy		X	X	X		
Acquisition strategy						X

larger transaction volumes, such innovations are currently constrained by varying levels of digital competence and capacity among SFSC stakeholders, such as producers and customers. To address these challenges, FH A and FH F are pursuing BMI through digitalization, though in different ways and to varying extents.

Digitalization has become central to FH A's innovation strategy. They are developing a digital platform that directly connects producers with consumers, transforming the value creation and delivery components of its BM. The platform is used to streamline business transactions and strengthen

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consumer relationships, expand distribution channels, and attract new customer segments through enhanced digital communication. Through the platform, producers are enabled to retain control over product offerings and pricing, while FH A complements the offering with marketing, logistics, and customer support services. This configuration strengthens producers' value capture through pricing autonomy and a larger share of sales revenue, while FH A generates service-based income and benefits from increased platform engagement. These digital activities represent a significant shift in how the organization interacts with its local community, introducing communication and outreach functions that were not previously part of its operations. Beyond transactional functions, FH A also employs its digital platform to extend its value proposition by promoting consumer education and knowledge exchange, particularly around production practices and producer identities. This educational emphasis differentiates FH A's offering, fosters trust and loyalty, and reinforces the perceived value of transparent and locally rooted food systems.

Established in southern Sweden during its start-up phase (2018–2023), FH A has since begun expanding toward larger urban markets with higher purchasing power. Although this move requires operating beyond the immediate geographic reach of its current producers, the long-term objective is to build new localized food systems in target cities by recruiting regional suppliers through the digital platform. This strategy enables geographical expansion of value creation and delivery while maintaining the integrity of FH A's local-food-based value proposition.

FH F is also pursuing digital BMI, primarily through the development of a digital platform aimed at optimizing local food sales and extending market reach. As this platform is still under development, its influence on value capture and the overall value proposition has yet to be realized. Nonetheless, both FH A and FH F illustrate how reconfiguring distribution channels through digitalization constitutes an important pathway for innovating their BMs.

4.2.2 | Diversification Strategy Affecting Distribution Channels

To varying degrees, FH A, B, and E have pursued diversification strategies aimed at broadening their distribution channels, resulting in distinct impacts across their BM components.

For FH A, incremental adaptations in the value proposition have centered on serving public procurement clients to ensure more stable, year-round revenue, affecting the way they capture value. This strategic shift broadens the customer portfolio from primarily private consumers toward institutional buyers, thereby reinforcing the reliability of income streams. Their existing resources are used to serve these customers; thus, no changes were implemented in the value creation and delivery of their BM.

Although FH E maintains financial stability, its value capture remains constrained by structural factors. It is dependent on public procurement contracts to provide a steady income, but experiences limited growth potential, as the tendering process is both competitive and resource intensive. To increase profitability and the chance to grow the business, FH E tried diversifying its customer portfolio by including consumers. This required changes in the value proposition, by reframing its value offering to suit consumers, as well as value creation and delivery, which was done through the development of a consumer-focused pop-up shop launched along a commuter route aimed at enhancing direct sales. Unfortunately, insufficient demand in combination with high competition from conventional retailers rendered the initiative economically unsustainable. Based on these experiences, FH E concluded that expanding its institutional customer base in neighboring regions represents the most viable avenue for scaling its value capture within existing resource constraints.

In contrast, FH B implemented more radical diversification measures that significantly reconfigured the value proposition, creating and delivery, and capture. By opening a restaurant adjacent to its retail shop, FH B strengthened its value proposition to consumers through an integrated offering of locally sourced products and prepared meals, thereby enhancing the consumer experience and reinforcing local food identities. The restaurant and retail shop together transform value creation and delivery by creating multiple, mutually reinforcing channels that promote regional producers and enable direct interaction with consumers. Additionally, this model supports circular practices by reducing food waste and generating broader environmental and societal benefits, aligning the value proposition with sustainability goals. To improve value capture, FH B responded to post-pandemic economic challenges by relocating to a smaller, energy-efficient facility adjoining its retail and restaurant operations. This relocation not only reduced operational costs and environmental impact but also increased storage efficiency, thereby enhancing profitability and resilience.

4.2.3 | Collaboration Strategy to Expand Market Reach

In recent years, FH B, C, and D have pursued a collaboration strategy that reshapes their value proposition by collectively seeking to expand market reach, particularly through the acquisition of larger HoReCa accounts. This joint approach enables the FHs to position themselves as reliable and professional suppliers capable of meeting the volume, consistency, and quality requirements of large-scale clients, thereby strengthening their overall market credibility and competitive advantage.

The strategy fundamentally reconfigures value creation and delivery through coordinated resource sharing, collective logistics, and joint market engagement within regional supply chains largely composed of local producers. While each FH continues to operate independently as a local wholesaler serving public procurement entities and HoReCa clients, their collaboration allows them to function as a networked system that delivers greater efficiency and scale than would be possible individually. By jointly targeting hotel and restaurant chains, the FHs can pool inventories, synchronize delivery schedules, and share marketing efforts, thereby improving service reliability and reducing duplication of effort.

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This cooperative model also enhances value capture by enabling economies of scale and reducing individual transaction costs, while simultaneously mitigating procurement risks for both the FHs and their customers. When one FH experiences temporary supply shortages, partner FHs can step in to provide supplementary products, ensuring supply continuity and customer satisfaction. This redundancy mechanism not only improves customer retention but also stabilizes revenue streams across the participating FHs.

4.2.4 | Acquisition Strategy to Expand Customer Base and Resources

To enhance its competitive position and support organizational growth, FH F has pursued an acquisition strategy by acquiring a conventional local wholesaler. This strategic move enhances the FH's value creation and delivery by broadening its customer base, expanding both tangible (infrastructure, logistics capacity) and intangible (supplier relationships, market knowledge) resources, and improving its ability to operate efficiently in a highly competitive market environment. Through the integration of the acquired wholesaler's infrastructure, client network, and supply chain relationships, FH F extends its delivery mechanisms beyond those originally developed, securing access to diversified and more stable distribution channels.

The acquisition simultaneously strengthens FH F's value capture by opening new market segments and increasing its bargaining power within local supply chains. These expanded commercial relationships enhance revenue stability and provide greater resilience against seasonal fluctuations in demand, which is particularly important in the tourism-dependent context in which FH operates.

The acquisition also reshapes FH F's value proposition. While maintaining its identity as a promoter of local, seasonal food, it now supplements this orientation with the reliability and scale associated with conventional wholesale operations. This dual positioning enables FH F to balance its sustainability-oriented mission with the economic imperatives of operating in a tourism-dependent market.

5 | Discussion

5.1 | BM Components of FHs

The results of this study showed that FHs' value intention tends to be multidimensional, with strong links to ideology. All FHs are led by ideologically driven owners (FH A–E) or operational managers (FH F) who cultivate professional relationships grounded in trust and mutual respect. While profitability remains a goal, the FHs have a large focus on shared-value creation between all actors in the SC (Berti and Mulligan 2016; Manikas et al. 2019; Stoeva et al. 2024). Deeply embedded in their local communities, FH managers position themselves as agents of change, striving for a fairer and more sustainable food system (Barth et al. 2017). In all cases, this stewardship mindset guides both operational decisions and BMI. The FH managers see themselves not merely as intermediaries but as caretakers of a symbiotic food ecosystem, where producers, distributors, and customers share responsibility for maintaining fairness, quality, trust, and sustainability. By aligning innovation with collective well-being, they translate stewardship values into concrete BM choices that balance profitability with care for stakeholders, regions, and future generations (Ulvenblad 2023).

The FHs balance commercial and mission-driven objectives, combining core operational activities such as aggregation, distribution, and marketing with broader commitments to producer support, community engagement, and sustainable food system development (Artoonian et al. 2023; Hyland and Macken-Walsh 2022; Manikas et al. 2019). These dual value propositions affecting both producers and customers were not merely rhetorical but were embedded in everyday practices, including prioritization of local sourcing, transparency, and community-oriented service design. With regard to operational models, the FHs examined corresponded to typologies identified in the literature, with two adopting a hybrid model and four oriented exclusively toward business-to-business customers (Hyland and Macken-Walsh 2022). Despite favorable consumer attitudes and policy support for local food (Regeringskansliet 2025), insufficient direct consumer purchasing limited financial sustainability, particularly for FHs with B2C components. Consequently, all FHs emphasized the critical role of large-volume sales to HoReCa and public procurement actors, highlighting the dependence on institutional markets for long-term resilience.

The analysis of value creation and delivery shows that the studied FHs draw on both tangible resources, such as logistics infrastructure and warehouses, and intangible resources, particularly human and relational capital (Hyland and Macken-Walsh 2022). Employees' sector-specific expertise, often rooted in prior experience as producers or within HoReCa and retail, proved critical in strengthening operational capacity and deepening ties with local food systems, extending earlier insights that primarily emphasized the role of FH owners (Barth et al. 2017). Consistent with scholars underscoring the need to align business and mission-driven objectives in FH BMs (Saul et al. 2021), the findings highlight that all FHs collaborated closely with small and medium-sized producers not only to handle core functions of sales, marketing, logistics, and distribution but also to deliver socially oriented services, including producer support, farm visits, and workshops. These practices underscore the dual role of FHs as both business facilitators and community actors, supporting prior research that stresses the importance of such partnerships for sustaining SFSCs (Berti and Mulligan 2016; Manikas et al. 2019; Tewari et al. 2018).

The findings indicate that FHs capture economic value primarily through sales markups, with one case also using producer subscription fees, while maintaining close collaboration and transparency with producers in line with SFSC principles (Manikas et al. 2019). Beyond financial returns, social value is also created for key stakeholders where customers and consumers benefit from fresh, locally sourced products with short lead times, producers gain fair prices that support reinvestment and welfare, and employees report loyalty and fulfillment tied to the mission-driven orientation of the FHs (Avetisyan and Ross 2019; Berti and Mulligan 2016). Additional benefits include potential

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rural development through job creation (Allen and Prosperi 2016) and, in some cases, reduced environmental impact via investments in sustainable distribution (Berti and Mulligan 2016).

5.2 | Strategies Contributing to FH'S Business Model Innovation

While the FHs investigated share many similarities in their BM components, their innovation strategies reveal noticeable differences. This reveals a central insight that although FHs appear homogeneous from the outside, their approaches to BMI differentiate them substantially in terms of market positioning and contribution to SFSC development.

Across the sample, FHs engaged in several types of BMI, each reflecting a distinct pathway through which BMs evolve (Geissdoerfer et al. 2018). BM startups, where the organization was created through a new BM, were present in the establishment of all FHs. Throughout the years, FH A, B, and E pursued BM diversification, developing additional BMs by expanding distribution channels and customer segments (Geissdoerfer et al. 2018). This allowed them to experiment with new value propositions and revenue streams without disrupting their core operations, thereby enhancing resilience.

Digitalization efforts by FH A and FH F illustrate BM transformation, in which existing BMs were modified to improve sustainability performance and strategic reach (Geissdoerfer et al. 2018). By developing digital platforms aimed at improving coordination, communication, and transparency, these FHs reshaped value creation and delivery while strengthening value propositions by promoting transparency, education, and closer producer–consumer relationships. Although implementation challenges remain, particularly related to digital capacity among producers, these initiatives expand opportunities for value capture through broader market reach, service diversification, and stronger customer engagement, positioning the FHs for greater scalability and long-term sustainability.

The collaboration strategy observed among several FHs represents another form of BM transformation. By jointly coordinating activities such as shared deliveries, the FHs reinforced their collective value creation and delivery across the network, while strengthening their value proposition and improving the potential for sustainable value capture through increased efficiency, reliability, and shared market access.

FH F's acquisition of a conventional wholesaler exemplifies BM acquisition, whereby an existing BM is identified, acquired, and integrated into the organization (Geissdoerfer et al. 2018). This strategy hybridized FH F's model by combining the mission-driven goals of SFSCs with the operational efficiency and scale associated with CFSCs. Through the integration of new resources, customer bases, and supply relationships, FH F enhanced its ability to create, deliver, and capture value, increasing its competitiveness in a geographically constrained and highly competitive market.

In examining these strategies through the lens of radical and incremental innovation (Boons and Lüdeke-Freund 2013), three FHs (A, B, and F) pursued radical innovation through digitalization, major diversification, or acquisition, while the remaining FHs (C, D, and E) adopted incremental changes such as increased collaboration and minor diversifications. Radical strategies were often motivated by entrepreneurial motivations to diversify their value delivery (FH B), a strategic reorganization of their customer base trying to overcome financial challenges (FH A), and the acquisition of a conventional wholesaler to withstand increased competition in a geographically limited market (FH F). By contrast, incremental BMI extends the FH's existing practices without fundamentally altering their role in the supply chain.

Despite FH owners' ambitions to drive systemic change in the agri-food landscape, many reported being constrained to incremental innovations due to size, limited resources, and market volatility. Value intentions also appeared to influence innovation pathways where FH A and B displayed reformist ambitions aligned with radical strategies, whereas FH F's acquisition reflected a shift away from mission-driven motivations toward more conventional market logics.

Finally, while some research points out that digitalization has emerged as a transformative driver of BMI in SFSCs, enabling the reshaping of practices and the optimization of operations (Csordás et al. 2022), the findings indicate that most FHs, as well as their producers and customers, exhibited low levels of digital maturity. Most FHs described that both producers and customers predominantly relied on telephone and email communication rather than digital ordering or booking systems. Implementing advanced digital technologies to optimize operations was therefore seen as challenging.

These innovation strategies raise critical questions in the context of alternative intermediaries in SFSCs. While radical strategies may enhance competitiveness, they risk eroding the mission-driven identity that underpins the FH's focus on trust, community embeddedness, and long-term sustainability. Equally, incremental approaches may safeguard values but struggle with scalability and financial resilience. This tension underscores a key dilemma: whether FHs must remain mission-driven to sustain SFSC ideals, or whether competitive pressures will inevitably shift some toward more conventional market logics, potentially diluting their transformative potential.

6 | Conclusions and Practical Implications

This study examined the BM and innovation strategies for BMI of six Swedish FHs using a comprehensive BM framework to deepen understanding of their role as alternative intermediaries in SFSCs. By analyzing all four value-related BM components, namely value intention, proposition, creation and delivery, and value capture, together with the innovation strategies shaping BMI, the study provides the first integrated overview of how FHs structure and develop their BMs. The findings show that while FHs share several common BM categories, they also exhibit context-specific elements that reflect differing local conditions and strategic priorities. Building on these insights, the study advances a conceptual FH BM framework (Figure 3) that captures both the core components and the primary innovation pathways

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through which FHs evolve. This framework demonstrates how value intention, proposition, creation, delivery, and capture are interdependent, and how innovation strategies, whether incremental or radical, shape FHs' capacity to adapt and support systemic change in SFSCs. Theoretically, the study contributes to research on sustainable SFSC development by linking BM components with the innovation strategies that drive BMI, offering a more nuanced understanding of how FHs operate, innovate, and potentially strengthen the sustainability and competitiveness of local food systems.

The findings of this study provide several practical implications for FH managers, policymakers, and other actors involved in SFSCs. The proposed BM framework can serve as a strategic tool for FH practitioners to assess and refine their operations by identifying which BM components offer the greatest potential for innovation. Managers can use the outlined innovation strategies to guide decision-making and adapt their BM to evolving market conditions and resource constraints.

The results also point to the need for capacity-building initiatives that strengthen managerial and digital competencies among FH operators and producers. Regional development agencies and support organizations can apply these insights to design training programs and funding schemes that facilitate innovation and collaboration among FHs. Furthermore, policymakers can use the conceptual framework to better align institutional support and policy incentives with the distinct value creation logics of FHs, thereby fostering sustainable and resilient food systems.

Finally, the study highlights in what way FHs have the capability of creating value within SFSCs. Unlike conventional intermediaries, FHs integrate social and environmental values into the core of their operations. They not only leverage tangible resources, such as infrastructure for distribution and marketing, but also place particular emphasis on intangible resources, including employee expertise and the cultivation of trust-based relationships with both producers and customers. As such, FHs emerge as unique actors that balance business and mission-driven objectives with operational effectiveness, contributing meaningfully to the sustainability and resilience of SFSCs.

7 | Limitations and Suggestions for Further Research

Some limitations should be noted. This study focused on FHs in Sweden, which may limit its generalizability. However, the six FHs included in this study operate in alignment with FH definitions in the literature; therefore, the results may still allow for analytical generalization beyond the specific cases. Future research should be conducted with FHs in other geographical, institutional, and policy environments to test the proposed FH BM framework and innovation strategies. Further, comparative studies between urban and rural regions or between different national food systems would also deepen understanding of the contextual factors shaping FH BMs and their capacity to adapt and innovate.

Quantitative assessments, such as analyses of interdependencies among BM components, performance indicators, or larger cross-hub comparisons, could also strengthen validation of the framework. Moreover, several of the strategies examined, particularly related to digitalization and collaboration, remain in early implementation stages. Longitudinal research is therefore needed to evaluate how these initiatives mature and influence BM evolution over time.

As digitalization increasingly reshapes food systems, there is a pressing need to examine its implications for the sustainability performance of FHs. Digital tools offer opportunities for logistical optimization, enhanced traceability, and more interactive consumer engagement, yet adoption remains limited due to resource constraints and technological hesitancy among small and medium-sized producers. Emerging technologies such as blockchain hold potential for transparency and trust-building, but their meaningful integration requires participatory innovation approaches that involve stakeholders in co-creation. Such approaches may enhance inclusivity, support shared value creation, and strengthen the scalability of SFSCs.

This research touched upon innovation strategies as a part of BM and BMI. The presented evidence suggests that while radical strategies can strengthen market positioning, they risk eroding the trust, community embeddedness, and shared-value orientation that underpin SFSCs. Incremental strategies, in contrast, safeguard these values but risk marginalization in competitive food markets. Future research should further explore this balance and investigate additional innovation strategies not covered here to understand how FHs can remain both viable and transformative actors within evolving food systems.

Finally, to support the strategic innovation of FHs, future studies could identify and analyze critical success factors, affected by internal and external processes, that enable the effective upscaling and institutionalization of FHs within regional and national food systems. This perspective is particularly important for positioning FHs as integral actors in sustainable food governance, where they can contribute to balancing economic viability with broader social and environmental objectives.

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